

Catalytic Pyrolysis of Cellulose Using Formulated Red Mud, Sand, and HZSM-5

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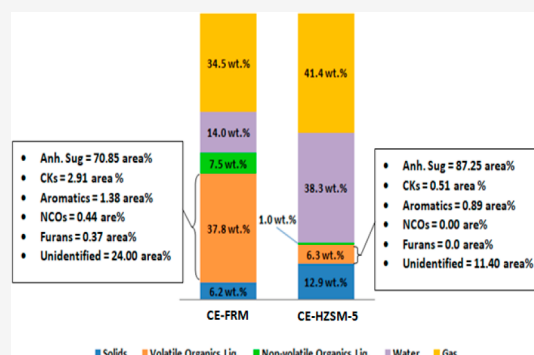


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Supporting Information

ABSTRACT: The catalytic pyrolysis of microcrystalline cellulose (CE) using formulated red mud (FRM) and HZSM-5 catalysts was conducted in a fluidized bed reactor at 400 °C. Both catalysts promoted the conversion of CE. The FRM promoted the formation of organic liquids and pyrolytic water, while the HZSM-5 catalyst promoted the pyrolytic water and the formation of pyrolysis noncondensable gases (PNCGs). The electrostatic precipitator (ESP) biooil (25.8 ± 2.54 wt %) from the CE-FRM consisted mainly of levoglucosan (LGA, 68.60 area %), cyclic ketones (CKs, 2.91 area%), and aromatics (1.38 area%), whereas the HZSM-5 biooil consisted of levoglucosan (82.27 area%). The major oxides of the FRM (i.e., Al₂O₃ and Fe₂O₃) were used during the pyrolysis of CE. Both catalysts promoted the formation of pyrolytic water and PNCGs. Compared to sand, the Fe₂O₃ promoted the formation of anhydrous sugars by about 6%, aromatics by more than 3-fold and CKs by more than 2-fold, whereas the Al₂O₃ promoted the formation of noncyclic oxygenates (NCOs) by more than 3-fold, furans by more than 3-fold, CKs by more than 5-fold, and aromatics by more than 28-fold. During the CE-FRM pyrolysis, larger particle size resulted in higher char/coke and pyrolytic water yields and lower organic liquid and PNCGs. This was due to the longer residence time of LGA in the larger particles where further dehydration and charring occur. The pyrolysis of D-glucose (DG) over the FRM catalyst resulted in higher char/coke and water yields and lower organic liquid and PNCGs compared to the catalytic pyrolysis of CE over FRM. Compared to the CE-FRM biooil, the DG-FRM biooil had higher furans content and lower yields of anhydrous sugars and aromatics. The longer residence time of LGA produced during the DG-FRM pyrolysis in the hot zone resulted in its further dehydration and polymerization to form char.



1. INTRODUCTION

Cellulose, a homopolymer of glucose, is the major component of the lignocellulosic biomass. Lignocellulosic biomass has been explored as an abundant and low-cost feedstock for biofuel production. Pyrolysis is one of the technologies that has received much attention to convert lignocellulosic biomass to biofuels and other products. Understanding the mechanisms involved during the pyrolysis of cellulose is necessary to identify the origin of some of the chemicals present in the lignocellulose pyrolysis biooil. Numerous studies investigated the catalytic pyrolysis of cellulose using two types of catalysts: (1) acidic catalysts such as zeolites,^{1–5} metal oxides,^{6–13} and carbonaceous acids^{14–18} and (2) basic catalysts such as alkali and alkaline earth metal oxides^{14,15,19–21} and metal chlorides.^{22–24} The acidic catalysts promoted the formation of aromatic compounds, which improved the fuel properties of the biooil. The basic catalyst promoted the formation of high-value chemicals that can be used as building blocks for other commercial chemicals.

Fast pyrolysis has been proven to be an effective technology that can convert biomass into biooil, biogas, and biochar.^{25,26} Catalytic fast pyrolysis (CFP) has been gaining more attention because its major product, i.e., biooil, can be coprocessed with

standard gas oil to produce fuel.²⁷ Furthermore, the CFP can promote the production of certain high-value chemicals such as LGO and LGA.^{6,28} Fluidized bed reactors allow good mixing of the feedstock and catalyst during catalytic fast pyrolysis, which allows good heat transfer into the feedstock particles that are fed into the reactor at a constant feeding rate. In a fluidized bed, attrition phenomena can result in the loss of catalyst particles that exit the reactor due to their reduced size.²⁹ Therefore, the catalyst used in such a reactor must maintain its structure at high temperatures.

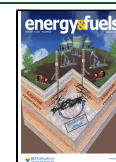
Biooil, the major product of the fast pyrolysis process, is a complex mixture of chemicals including aldehydes, esters, acids, ketones, anhydrosugars such as levoglucosan (LGA) and levoglucosenone (LGO), and other oxygenated molecules. The wide range of chemicals in the pyrolysis biooil makes it

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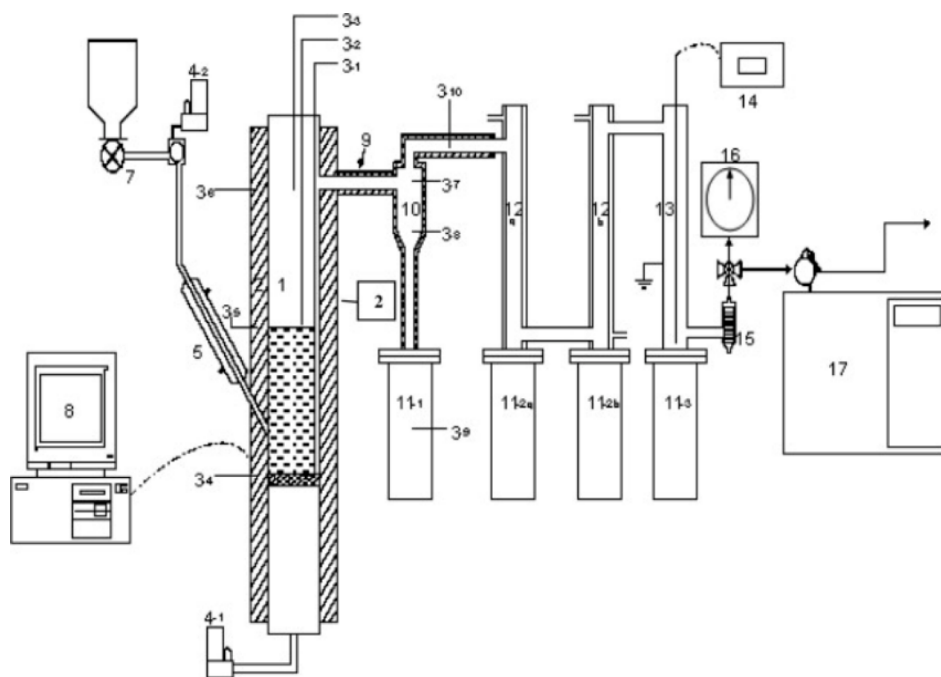


Figure 1. Laboratory bench-scale pyrolysis unit:⁴⁵ (1) fluidized bed reactor, (2) heating element, (3) thermocouples, (4) gas flow rate controller, (5) feedstock feeding tube, (6) feedstock hopper, (7) feedstock screw feeder, (8) pyrolysis unit controller, (9) insulated reactor outlet, (10) hot gas filter, (11) condenser reservoir, (12) condenser, (13) electrostatic precipitator, (14) power supply, (15) coalescing filter, (16) gas flow meter/totalizer, and (17) gas chromatograph.

challenging to economically separate high-value chemicals such as LGA and LGO. Selective pyrolysis was proven to promote the production of certain target high value via the use of the catalyst or the pretreatment of the pyrolysis feedstock.^{6,30–32}

All of the aforementioned catalysts can be costly and create unwanted hazardous waste, which can be a nuisance to the environment. Such drawbacks can lower the profit margin of the pyrolysis process. Therefore, it is important to identify catalysts that are cheap and recyclable. Red mud is the waste product of the Bayer process for aluminum mining. It is a mixture of oxides including, but not limited to, Fe_2O_3 , Al_2O_3 , CaO , TiO_2 , NaOH , etc. The red mud received a lot of attention as a catalyst for the pyrolysis process.^{33–37} Yang et al. conducted the pyrolysis of cellulose using red mud-supported transition-metal catalysts at 600 °C. They reported that the cobalt/red mud catalyst significantly promoted the production of furans, whereas the addition of Mo, Cu, and Ni to the red mud support promoted the formation of furans and aromatics, ketones, and aromatics, respectively.³⁸ Shao et al. reported an effective method for the production of ketones and aldehydes from the catalytic pyrolysis of holocellulose at 400 °C using red mud composite catalysts consisting of ZrO_2 loaded onto the red mud.³⁹ Wang et al. conducted the catalytic pyrolysis of corn stover using red mud and its major oxides as catalysts at 500 °C. They reported that Fe_2O_3 promoted the production of acids and furans, whereas CaO promoted the formation of ketones. On the other hand, Al_2O_3 , TiO_2 , and SiO_2 promoted the production of acids and inhibited the formation of methoxyphenols.⁴⁰ Li et al. conducted the catalytic pyrolysis of rape straw using red mud in tandem with HZSM-5 at 550 °C. They reported that the use of the RM followed by HZSM-5 promoted the coking resistance of the HZSM-5 and the cascade dual-catalyst system consisting of acid-treated RM and HZSM-5 produced 12.91% of benzene, toluene, and xylene.⁴¹

Agblevor and Yathavan studied the red mud-catalyzed pyrolysis of pinyon-juniper biomass in a fluidized bed reactor.²⁹ They reported that the physical structure of the red mud catalyst as is was proven to be weak.²⁹ One problem associated with the use of red mud in a fluidized bed reactor is the loss of the red mud catalyst, which exited the reactor due to attrition phenomena and the fluidizing nitrogen gas. To address such problem, Agblevor et al. created the formulated red mud (FRM) by adding binders (colloidal alumina and colloidal silica) to the raw red mud. The FRM consisted of 60 wt % red mud, 20 wt % colloidal alumina, and 20 wt % colloidal silica⁴² and was structurally stronger than the raw red mud. The FRM was used for the catalytic pyrolysis of pinyon-juniper (PJ) biomass,⁴² and the biooil produced had similar viscosity to that produced from the red mud-catalyzed pyrolysis of PJ.⁴³ Furthermore, no significant loss of activity catalytic was observed after several regenerations of the FRM catalyst.³⁸ Due to its stable catalytic activity and resistance to poisoning by alkali and alkaline earth metals from the ash in biomass, the FRM is a potential catalyst for the selective pyrolysis of lignocellulosic biomass. Because of the complexity of lignocellulosic biomass, it is necessary to study its major polymers separately to understand their contribution to the pyrolysis products yields and the quality of the biooil.

The purpose of this study is to investigate the effect of FRM on the CE pyrolysis product distribution and biooil composition. HZSM-5 was used for comparison purposes. The results from the pyrolysis of CE using sand were used as a baseline. Furthermore, the effect of cellulose particle size as well as the major oxides of FRM will be investigated.

2. MATERIALS AND METHODS

2.1. Materials. The microcrystalline cellulose used in this study was Avicel PH-101 with an average particle size of 50 μm (MilliporeSigma, Burlington, MA). The proximate analysis of the

cellulose was performed using a TGA Q500 (TA Instruments, New Castle, DE). 20 mg of cellulose was heated from room temperature to 575 °C at 15 °C/min under nitrogen gas. The sample was held at 575 °C for 10 min, then the carrier gas was switched from nitrogen to air, and the temperature was held for 10 min. The ultimate analysis of the cellulose was performed using a Flash 2000 CHNS-O elemental analyzer (Thermo Fisher, Waltham, MA). About 2 mg of cellulose was loaded into a tin capsule and then combusted in the reactor, where the gases were reduced and analyzed using a thermal conductivity detector (TCD) to determine the nitrogen, carbon, hydrogen, and sulfur in the sample. The oxygen content was calculated by difference after accounting for the C, N, H, and S. DG and was purchased from Fisher Scientific (Fair Lawn, NJ). Cellet700 (CE700) was received from Glatt Air Techniques, Inc. (Ramsey, NJ). The CE700 consisted of microcrystalline cellulose with a particle diameter of 0.7–1 mm.

2.2. Catalysts. The FRM and HZSM-5 catalysts were from the same batches used in our previous study.⁴⁴ The protonated ZSMS, i.e., HZSM-5, was obtained by calcining the former at 550 °C for 5 h. The FRM, HZSM-5, and sand were dried overnight at 105 °C and then sieved to separate the particles in the range 125–180 μm .

2.3. Thermogravimetric Analysis of Pure Cellulose and Cellulose–Catalyst Mixtures. The thermogravimetric analysis of pure CE, as well as the mixtures of CE–sand, CE–FRM (1:1), and CE–HZSM-5 (1:1), was performed using a TGA Q500 thermogravimetric analyzer (TA Instruments, New Castle, DE). About 20 mg of CE or CE–medium mixture was loaded into a platinum pan and then heated from room temperature to 700 °C at a heating rate of 15 °C/min under 100 mL/min of ultrahigh-purity nitrogen gas. The weight loss was recorded as the sample was being heated.

2.4. Surface Area Analysis. The surface area of the FRM and HZSM-5 catalysts was conducted using an MS-16 Quantachrome BET surface analyzer (Quantachrome Instruments, Boyton Beach, FL). About 0.2–0.3 g of the sample/standard was degassed for about 3 h at 300 °C. The standard used was an Anton Paar QuantaTec surface area reference material (Anton Parr QuantaTec Inc., Boyton Beach, FL, Cat. No. 193441).

2.5. Pyrolysis Experiments. The pyrolysis experiments were performed on a laboratory bench-scale reactor. The different components of the pyrolysis unit are shown in Figure 1. The pyrolysis unit consists of a feedstock hopper, where the cellulose was loaded and then fed into a bubbling fluidized bed reactor through the combined action of a twin screw feeder and nitrogen gas. The fluidized bed consists of 100 g of sand, FRM, or HZSM-5 that is suspended using nitrogen gas. After the CE is fed into the reactor, the pyrolysis vapors and fine particles of the solids are separated by the hot gas filter. The pyrolysis vapors are then quenched in the first and second condensers. A mixture of 50–50 water and ethylene glycol was used to maintain the two condensers at –8 °C. The aerosols and any pyrolysis gases that escape both condensers were captured by the electrostatic precipitator (ESP) kept at –30 kV followed by a coalescing filter. The noncondensable gases were collected in Teldar Bags Push Lock Valve, 1 L gas bags (SUPELCO, Belford, PA) and then analyzed using an SRI 8610C multigas #3 gas chromatograph (GC) (SRI Instruments, Torrance, CA). The GC was equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). The gases were separated using two GC columns, i.e., a 6 in. HySep D column and a 6 in. Mol Sieve 13X column. The biooil and solid residue were determined gravimetrically by weighing each component of the pyrolysis unit before and after the experiment. The PNCG values were calculated by difference.

The minimum fluidization velocity was determined experimentally for sand (particle size: 125–180 μm), FRM (particle size: 125–180 μm), and HZSM-5 (particle size: 125–180 μm). About 100 g of the medium was loaded into a glass reactor, and then nitrogen gas was used to fluidize the medium bed. The pressure at the bottom and at the top of the medium bed was recorded at different flow rates, and the pressure drop was plotted against the superficial velocity (Figure S3a–c in the Supporting Information). The minimum fluidization

velocity of the FRM and HZSM-5 bed was about 2.0 cm/s (2.4 L/min at 25 °C) compared to 4.1 cm/s (5 L/min at 25 °C) for sand.

The reaction temperature was set to 400 °C. The total gas flow rate consisted of (1) 4.5 L/min of nitrogen going through the reactor feeding tube to push the feedstock (CE) into the reactor and prevent the backflow and condensation of the pyrolysis vapors inside the feeding arm and (2) the fluidizing gas coming from the bottom of the reactor and through the catalyst/sand bed at 2 times the minimum fluidization velocity. All pyrolysis experiments were run in duplicates.

2.6. Biooil Analysis. **2.6.1. Water Content of the Biooil.** The water percentage in the biooil samples was quantified by volumetric titration using a Karl Fisher titrator (Brinkmann Instruments, Inc., Westbury, NY). The titrator consisted of two components: (1) a Metrohm 701 KF Titrino connected to (2) a 703 titration stand. Hydranal composite-5 was used as the titration reagent. The titration experiment was conducted as follows: about 0.02–0.03 g of the liquid sample (biooil) was injected into the titration glass pot containing about 50 mL of methanol. The titration started when the Hydranal composite-5 was gradually added by an automatic pump until all water in the biooil sample reacted with the reagent. The water content of the sample was then displayed as weight percent.

2.6.2. GC-MS Analysis of the ESP Biooils. The analysis was performed using an Agilent Technologies 7890A GC instrument with a Restek Rtx-1701 column (60 m $\times$ 250 μm $\times$ 0.25 μm) and an Agilent 5975C MS instrument with the NIST 2.0f Mass Spectral Search Program. Methanol was used as the solvent to dissolve the ESP biooil samples. The GC method used was as follows: helium flow rate 1 mL/min, injection volume 1 μL , injection port temperature 250 °C, split ratio 30:1, and initial oven temperature 45 °C (10 min) ramped at 3 °C/min to 250 °C (5 min). The mass spectrometer parameters were: ion source temperature was 230 °C, the solvent cut time was 6.5 min, the ionization energy was 70 eV, and the analysis was performed in scan mode over a scan range of 40–500 m/z . The data were processed using the Enhanced Chemstation software. The compounds reported in this study were those that matched one of the compounds in the software library with a percent match of 70% and higher (quality % in Tables S5–S11). The GC/MS analysis was done for samples that were prepared using the same mass of biooil from the FRM, HZSM-5, and sand experiments, and about the same amount of solvent was added to each sample to dissolve it; therefore, it was reasonable for us to do a relative comparison of the different samples. It is worth mentioning that the GC/MS analysis of the biooil samples was qualitative, not quantitative. This was due to the lack of calibration curves for the compounds identified by the software.

3. RESULTS AND DISCUSSION

3.1. Catalysts. The specific surface areas of the FRM and HZSM-5 were 98 and 131 m^2/g , respectively. Agblevor et al. reported that the FRM has an acidity of 0.095 mmol/g, a pore size of 10 Å, and a pore volume of 0.3 cm^3/g .⁴² The acidity of the HZSM-5 was 2.39 mmol/g.⁴⁶ The pore size and pore volume of HZSM-5 were not measured in this study. The HZSM-5 acidity was significantly higher than that of the FRM, whereas the pore size of the FRM catalyst was significantly higher than the well-known HZSM-5 pore size (5–6.36 Å).^{47–49} LGA is the major decomposition product during cellulose pyrolysis. The kinetic diameter of LGA is about 6.7 Å,⁴⁷ which is smaller than the average pore diameter of the FRM catalyst. Therefore, it was plausible to assume that the LGA molecules can be converted on the surface of the FRM and inside its pores. In contrast, the kinetic diameter of an LGA molecule is larger than the average pore size reported for HZSM-5; therefore, it was believed that the LGA conversion was more likely to take place on its surface but not as likely inside its pores.

3.2. Thermogravimetric Analysis. The thermograms of CE, CE–sand, CE–FRM, and CE–HZSM-5 are listed in

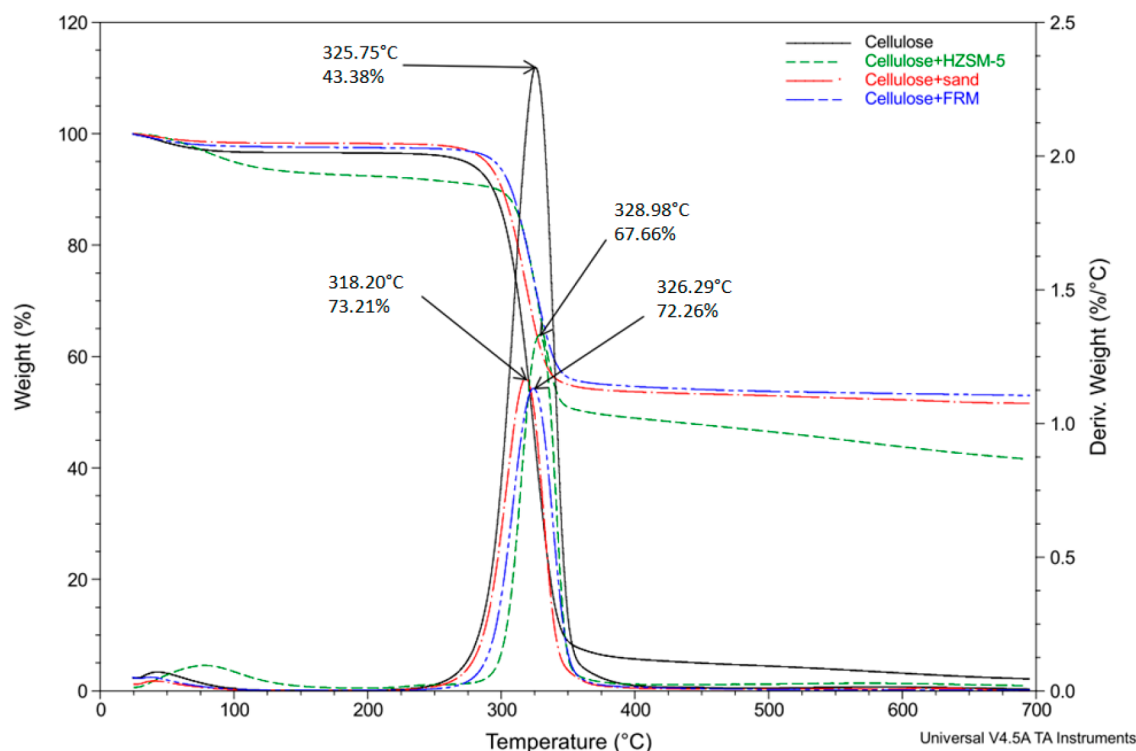


Figure 2. TGA of CE, CE mixed with sand, FRM, and HZSM-5.

Figure 2. The DTG peak of CE and that of CE–sand were between 250 and 400 °C, which was similar to what was reported in the literature for CE.^{50,51} The temperature of the maximum weight loss of the CE (325.75 °C) was slightly higher than that of the CE–sand (318.20 °C). This was because the addition of sand to the CE improved the heat transfer; hence, the maximum of the thermal decomposition of the CE occurred at lower temperature than that of the pure CE. The weight loss of the CE–sand sample started slightly earlier than that of the CE–FRM and CE–HZSM-5. The maxima of the DTG peaks were in the order CE–sand < CE–FRM < CE–HZSM-5. It is worth mentioning that the final weight of the CE–HZSM-5 was lower than that of the CE–sand and CE–FRM, which was probably due to the initial weight loss of the physisorbed water on the HZSM-5. The effect of red mud on the TGA curve of CE was in agreement with the study of Yang et al.⁵²

3.3. Catalytic Pyrolysis of CE: Effect of the Pyrolysis Medium. The distribution of pyrolysis products from the pyrolysis of CE using sand, FRM, and HZSM-5 is shown in Figure 3 and Table S1. The major pyrolysis product for the noncatalytic pyrolysis was the organic liquid product (43.0 wt %). However, our liquid product yield was much lower than those reported in the literature (close to 80 wt %).^{53,54} The reason for this difference was attributed to the fact that the high pyrolysis liquid yields were obtained using a pyroprobe or a microreactor, where the small sample size used (2–5 mg) ensures a good heat transfer. Furthermore, the pyrolysis vapors residence time in pyroprobe and microreactors was about 0.5 s compared to 4.7 s in our fluidized bed reactor. It is well understood that the longer residence time of the pyrolysis vapors in the reactor promotes their secondary reactions into char and PNCGs.^{55–57}

As for the catalytic pyrolysis, the major pyrolysis products were the organic liquids (45.3 wt %) for the FRM-catalyzed

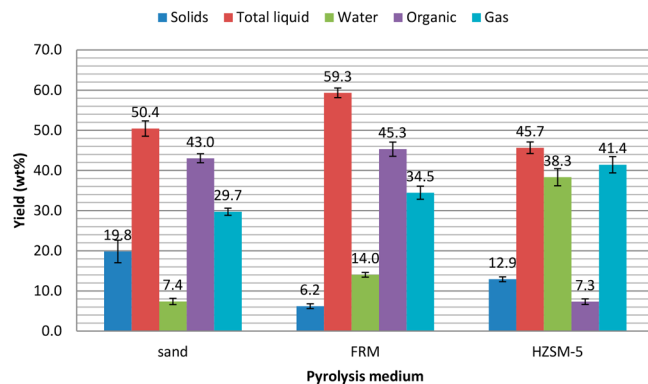


Figure 3. Pyrolysis product distribution from the CE pyrolysis using sand, FRM, and HZSM-5 at 400 °C.

pyrolysis and pyrolytic noncondensable gases (PNCG) (41.4 wt %) for the HZSM-5 catalyzed pyrolysis. Similar to the HZSM-5 catalyst, the FRM-catalyzed pyrolysis produced more PNCG (34.5 wt %) than the noncatalytic one (29.7 wt %). The catalytic pyrolysis of CE using acidic catalysts,^{1–5,58} metal oxides,^{7–13} and basic catalysts such as alkali and alkaline earth metal oxide^{19,19–21,59} is well-known to produce more PNCG than the nonpyrolytic pyrolysis due to catalytic cracking of the pyrolysis vapors. Compared to the noncatalytic pyrolysis, the FRM produced 5.3% more organic liquid, whereas the HZSM-5 catalyst produced 83.0% less organic liquids. Zeolites are acidic catalysts known for promoting the cracking of the pyrolysis vapors into PNCGs.^{1–5,58} The HZSM-5 catalyst used in this study has an acidity of 2.39 mmol/g, whereas the FRM has an acidity of 0.095 mmol/g; therefore, it was expected that the FRM would not be as effective as the HZSM-5 in cracking the pyrolysis vapors. On the other hand, transition metals and metal oxides are known to promote the production of

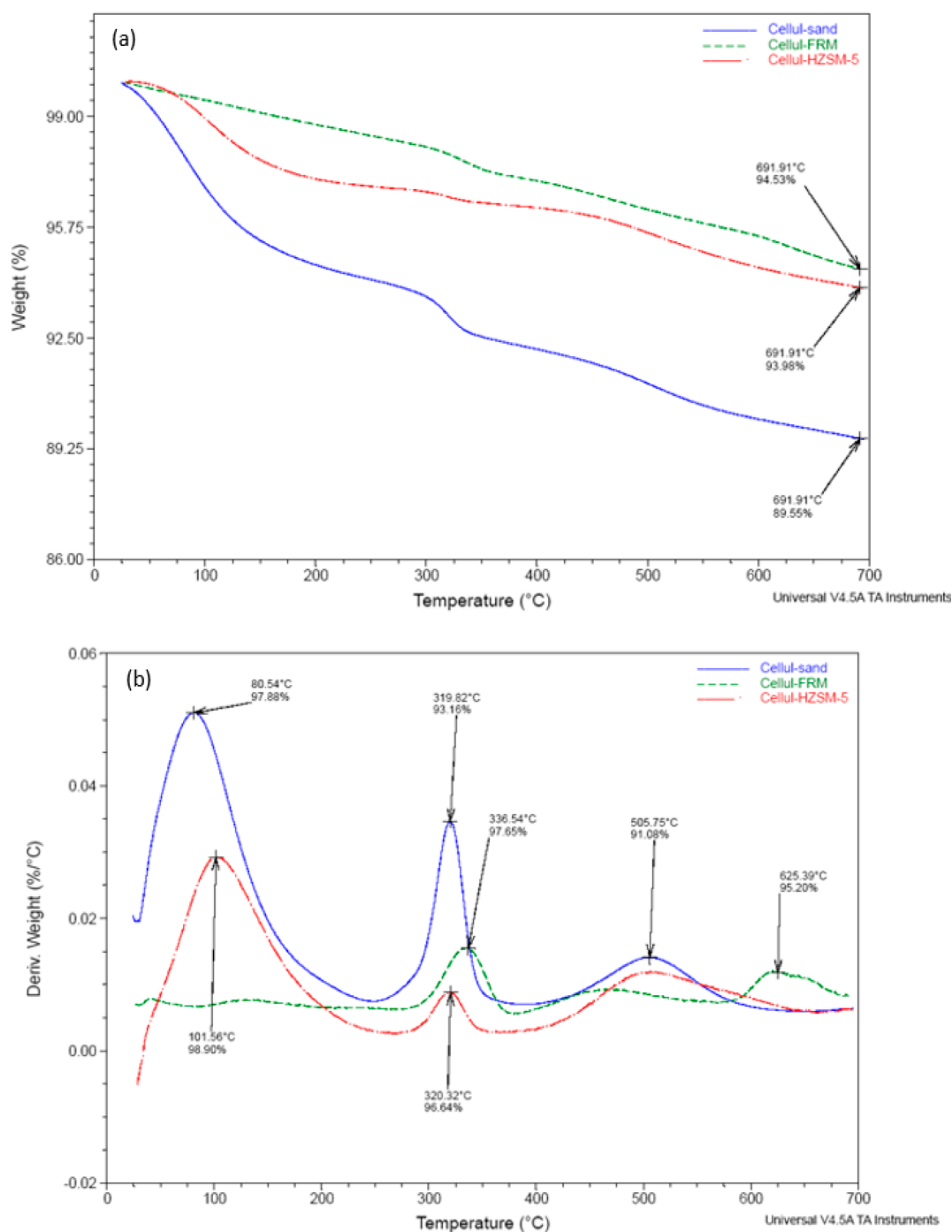


Figure 4. TGA (a) and DTG (b) curves of the solid residues from the pyrolysis of CE using sand, FRM, and HZSM-5.

levoglucosan during the catalytic pyrolysis of CE;^{60–62} therefore, it was believed that the higher organic liquid yield from the FRM-catalyzed pyrolysis of CE was due to an increase in LGA production.

The pyrolytic water content of the noncatalytic pyrolysis of CE was 7.4 wt % compared to 14 and 38.3 wt % for the CE–FRM and CE–HZSM-5 experiments. It is worth mentioning that while the use of FRM increased the water production by about 2-fold, the HZSM-5 produced about 5-fold more water compared to the CE–sand experiment. These results indicated that HZSM-5 was more efficient than the FRM in promoting the dehydration reaction. In fact, the HZSM-5 is well-known for promoting the production of aromatic compounds from the

pyrolysis of CE.^{63–65} This process consists of four steps; first, the LGA is formed by dehydration of CE, and then, the HZSM-5 catalyzes the formation of furans from LGA via dehydration reactions. Then, the furans are broken down into olefins, CO, CO₂, and water, and finally, the olefins undergo oligomerization, cyclization, and hydrogen transfer reactions, which are catalyzed by the HZSM-5 catalyst, to give aromatic compounds.⁶⁶

During the noncatalytic pyrolysis of CE, the char/coke yield was 19.8 wt %, whereas the solid yield from the FRM-catalyzed pyrolysis was 6.2 wt %. It was believed that the reactive oxides in FRM promoted the decomposition of the CE, resulting in a lower solid residue and more liquid and PNCG products. In

fact, iron has been reported to weaken the hydrogen bonds in the crystalline CE which led to its further decomposition during pyrolysis.⁶⁷ Additionally, alkali and alkaline earth metal are well-known to promote dehydration, rearrangement, and fragmentation reactions.⁶⁸ In contrast, the HZSM-5-catalyzed pyrolysis of CE produced less char/coke (12.9 wt %) and more water and PNCGs than the noncatalytic pyrolysis, which indicated that the HZSM-5 also promoted the decomposition of CE during pyrolysis. In order to confirm that the FRM and HZSM-5 promoted the decomposition of CE during pyrolysis, the TGA and DTG of the char/coke from the CE–sand, CE–FRM, and CE–HZSM-5 experiments were conducted, and the results are shown in Figure 4.

The TGA curves showed that the char/coke from the CE–sand pyrolysis exhibited a greater weight loss than that from the CE–FRM and CE–HZSM-5 pyrolysis. The DTG peak between 250 and 400 °C (CE decomposition range) indicated greater weight loss for the CE–sand pyrolysis compared to that of the CE–FRM and CE–HZSM-5. Furthermore, a third weight loss DTG peak (375–600 °C) was observed for the noncatalytic and catalytic pyrolysis solid residues and was attributed to secondary char decomposition. This peak was centered at about 505 °C for the sand and HZSM-5 samples and 450 °C for the FRM sample. This suggested that the secondary char formed during the FRM-catalyzed pyrolysis was less stable compared to the one produced from the noncatalytic pyrolysis and HZSM-5-catalyzed pyrolysis. The TGA/DTG curves confirmed that the FRM as well as the HZSM-5 promoted CE decomposition compared to sand. The HZSM-5 catalytic solid residue showed the smallest DTG peak for residual CE, which meant that HZSM-5 was more effective than the FRM and sand in converting the CE. However, the solid residue from the HZSM-5-catalyzed pyrolysis was higher than that of the FRM-catalyzed pyrolysis. This suggested that HZSM-5 promoted the formation of secondary char formation, which was corroborated by the significantly higher water yield from the CE–HZSM-5 experiment compared to the CE–sand and CE–FRM experiments. In fact, dehydration of CE promotes the charring reactions during CE pyrolysis.⁵⁵

Compared with sand, the HZSM-5 catalyst hindered the char/coke and organic liquid yields and promoted the pyrolytic water and PNCG yields. It has been established that the pyrolysis of CE using zeolite promoted the dehydration of LGA into furans, which can be further dehydrated and decarboxylated into olefins. Subsequently, the olefins can undergo cyclization to produce aromatic compounds. On the other hand, the use of zeolites during the catalytic pyrolysis of CE resulted in an increase of the PNCG product, which was due to the decarboxylation, decarbonylation, and cracking reaction that are promoted by zeolite catalysts.^{3,69,70}

The pyrolytic gas product composition of the noncatalytic and catalytic pyrolysis of CE using FRM and HZSM-5 is shown in Table S1. The noncatalytic pyrolysis gas product consisted mainly of CO and CO₂ (34.3% and 43.4%, respectively), which was in agreement with the literature.^{71–74} Ansari et al. reported that while the CO was due to the decomposition of furans, the CO₂ was due to the decomposition of pyrans/anhydrous sugars.⁷¹ Methane (5.1 mol %), gaseous hydrocarbons such as C₂H₆, C₃H₈, C₄H₁₀, and C₆H₁₄ (total = 1.6 mol %), and hydrogen (9.3 mol %) were also detected. The FRM-catalyzed pyrolysis promoted the production of CO₂ by 67.8% and decreased the production of CO by 11.7%. Red mud has been reported to produce more

CO₂ than CO during the pyrolysis of CE⁵² and biomass.^{29,75} This was believed to be due to the fact that the lattice oxygen in the Fe₂O₃ of the FRM participated in the pyrolysis reaction by migrating into the liquid and gas products, which resulted in the increase of the CO₂ as well as the water yields.⁷⁶ Furthermore, CaO has been reported to promote ketonization on its surface as well as the formation of acetate, which can decompose into acetone, CO₂, reactions formation of CO₂, acetone, and water.^{77,78} In contrast, the HZSM-5-catalyzed pyrolysis of CE promoted the production of CO by 53.4% and decreased the production of CO₂ by 1.9%. In fact, it has been reported that the acid sites of HZSM-5 promoted decarbonylation over decarboxylation during pyrolysis.^{79–82} It is worth mentioning that the CO yield was correlated to the aromatics yield when HZSM-5 catalysts with different acidity were tested during pyrolysis.⁸³

The gaseous hydrocarbon (C₂–C₆) production increased by more than 3- and 2-fold when FRM and HZSM-5 were used, respectively. On the other hand, the hydrogen content increased by 28.9% during the CE–FRM pyrolysis and decreased by 48.3% during the CE–HZSM-5 pyrolysis. Clearly, the CE pyrolysis using FRM and HZSM-5 involves different reaction pathways, which are also different from the noncatalytic ones. In fact, Razaei et al. reported that acid catalysts promoted the dehydration of the CE pyrolysis vapors to form furans. Simultaneously, the acid sites of the catalyst promote the hydrogen transfer to reactive species in the pyrolysis vapors, leading to the formation of carbocations that can further react to form olefins (C₂–C₆) via β -hydrogen elimination.⁸⁴ Furthermore, Jae et al. reported that the dehydration of LGA can produce molecules that are smaller than the ZSM-5 pores.^{73,85} Once these molecules diffuse into the ZSM-5 pores, they undergo a series of reactions to produce aromatics.⁸⁶

3.4. Catalytic Pyrolysis of CE: Effect of CE Particle Size. The pyrolysis products yields as well as the PNCGs product composition from the FRM-catalyzed pyrolysis of CE700 along side the data from the CE are shown in Figure 5 and Table S2. The catalytic pyrolysis of CE700 produced more char and pyrolytic water and less PNCGs and organic liquids than the CE. It is well-known that the particle size can significantly affect the pyrolysis product distribution due to mass and heat transfer effects.⁸⁷ Compared to CE, the char/coke yield as well as the water yield from CE700–FRM

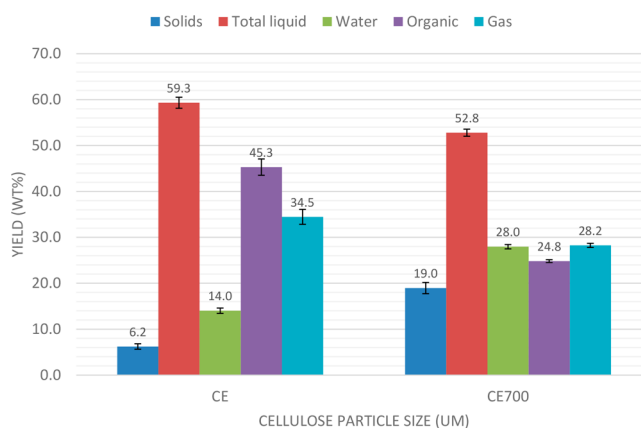


Figure 5. Effect of CE particle size on the pyrolysis product distribution from the FRM catalyzed pyrolysis of CE at 400 °C.

increased by 206.5% and 100.0%, respectively. This was believed to be the result of poor heat transfer due to the size of the CE700 particles (700–100 μm). We speculated that the pyrolysis vapors generated inside the CE700 particles spend a longer time inside the hot particles before reaching its outer surface and exiting the reactor. The longer residence time of the pyrolysis vapors inside the hot particles led to their polymerization and charring, which was corroborated by the higher pyrolytic water yield from the CE700 experiment.⁵⁵ Furthermore, it has been established that the char-catalyzed dehydration of levoglucosan leads to the formation of secondary char.⁸⁸ In fact, the thermal polymerization of LGA occurs above 250 $^{\circ}\text{C}$.⁸⁹

The composition of PNCGs from the pyrolysis of both CE particle sizes is shown in Table 2. As the CE particle size increased, the CO_2 yield increased by 74.9%, whereas the CO yield increased by 135.6% as the CE particle size increased. The increase of the CO yield was likely due to the decomposition/charring of LGA.⁸⁸ In fact, Guo et al. reported that compared to CO_2 , it was theoretically easier to produce CO from LGA.⁹⁰ Furthermore, they reported that the dehydration reactions of LGA facilitated the subsequent production of CO and inhibited the formation of CO_2 . The CE700 promoted the formation of CH_4 and hindered the production of the $\text{C}_2\text{--C}_6$ hydrocarbons, acetaldehyde, and acetone compared to the CE.

3.5. Effect of Al_2O_3 and Fe_2O_3 on the CE Pyrolysis Products Yields. The pyrolysis product distribution from the catalytic pyrolysis of CE using the two major oxides in FRM (Al_2O_3 and Fe_2O_3) is shown in Figure 6. Compared to the

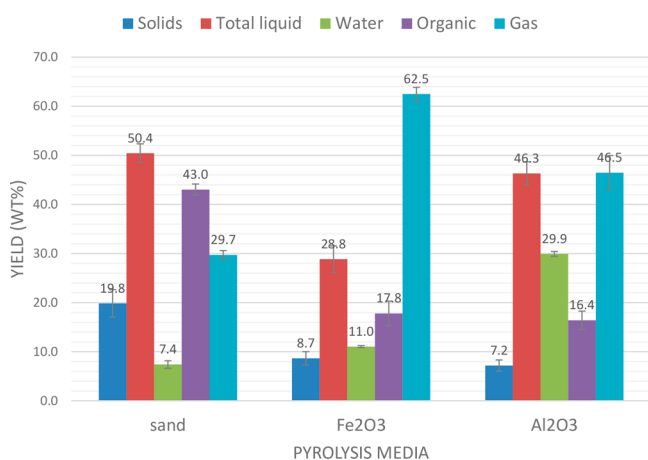


Figure 6. Pyrolysis product distribution from the pyrolysis of CE using sand, ferric oxide, and aluminum oxide at 400 $^{\circ}\text{C}$.

sand, the ferric oxide decreased the char/coke and organic liquids yield by 56.1% and 58.6%, respectively. On the other hand, the ferric oxide catalyst promoted the production of the pyrolytic water and the PNCGs by 48.6% and 110.4%, respectively. Therefore, Fe_2O_3 promoted the dehydration of CE and its depolymerization into PNCGs and hindered the formation of char/coke. Dobelet et al. reported that iron catalyzed the depolymerization of CE and the cleavage of its glycosidic bonds.⁶² In contrast, Collard et al. reported that during the pyrolysis of CE, iron promoted the formation of water, and PNCGs, while decreasing the tar yield compared to noncatalytic pyrolysis.⁹¹ They also reported that iron promoted the formation of char during the pyrolysis of CE,

which was the opposite of what we observed. This was probably because the fluidized bed reactor offered better heat transfer than the fixed bed reactor. Therefore, the cellulose conversion was more efficient in our fluidized bed reactor, which resulted in a lower char/coke yield and higher PNCGs. The high char/coke yield obtained by Collard et al. could also be the result of the higher reaction temperature (600 $^{\circ}\text{C}$) and the longer vapor residence time (12 s) in their tubular reactor compared to our fluidized bed reactor (4.7 s at 400 $^{\circ}\text{C}$). The longer residence time and the high reaction temperature could have promoted the polymerization and charring of the pyrolysis vapors.⁵⁵

Collard et al. reported that upon the impregnation of CE with Fe and Ni, the CO yield in the PNCGs decreased, whereas the yields of CO_2 and H_2O increased.⁹¹ They also reported that these results were due to the ability of metals (particularly the transition metals) to catalyze the dehydration reactions prior to the depolymerization of CE.^{92–94}

The use of alumina as a catalyst during the pyrolysis of CE resulted in a similar trend of the pyrolysis products as that observed when Fe_2O_3 was used. The alumina promoted the production of pyrolytic water and PNCGs by 304.1% and 56.6%, respectively, and hindered the production of char/coke and the organic liquids by 63.6% and 61.9%, respectively. Al_2O_3 is known to promote the dehydration reactions.^{95,96} Additionally, the Lewis acid sites in alumina promoted the formation of furans through the dehydration of sugars.⁹⁷ Furthermore, alumina has been reported to promote the cracking of high-molecular-weight compounds into lower-molecular-weight compounds and noncondensable gases.⁹⁸

Both the alumina and the ferric oxide decreased the total liquid and char/coke yields and promoted the production of the pyrolytic water as well as the PNCGs. These results indicated that both catalysts had superior activity compared with the sand. In fact, Gupta et al. reported that Al_2O_3 , Fe_2O_3 , and TiO_2 promoted the production of PNCGs during the catalytic pyrolysis of corn stover. They also emphasized on the effect of the Fe_2O_3 catalyst, which promoted the cracking of large organic molecules into PNCGs and reduced the liquid and char yields.⁹⁹ It is worth mentioning that the results of the CE pyrolysis using iron oxide and alumina may vary under an alkaline environment. In fact, the sodium in red mud is known to promote the dehydration reactions as well as carbon–carbon bond cleavage.⁴⁴ Furthermore, the study of the synergistic effect of the different oxides (Al_2O_3 , Fe_2O_3 , SiO_2 , TiO_2 , and Na_2O) of the red mud is needed to better understand the catalytic activity of the red mud during pyrolysis.

The PNCGs product distribution is shown in Table S3. The major gas product for the Al_2O_3 catalyst was CO (45.3 mol %), whereas CO_2 (46.9 mol %) was the major gas product when Fe_2O_3 was used. Compared to the CE–sand experiment, the alumina catalyst produced 106.2% more CO_2 and 120.8% more CO. Furthermore, the alumina produced no hydrogen gas, whereas the sand produced 9.3 mol % hydrogen. Compared to the sand, the hydrocarbon gas ($\text{C}_2\text{--C}_6$) content decreased by 36.6% when alumina was used and decreased by 12-fold when ferric oxide catalyst was used. Both Al_2O_3 and Fe_2O_3 catalysts promoted the production of acetaldehyde by more than 6-fold and more than 2-fold, respectively. As for acetone in the PNCGs, the Al_2O_3 and Fe_2O_3 catalysts promoted its production by more than 3-fold and more than 2-fold, respectively.

3.6. CE vs D-Glucose Pyrolysis Using FRM Catalyst.

The pyrolysis product distribution of DG using FRM is shown in Figure 7. The DG produced more char/coke and pyrolytic

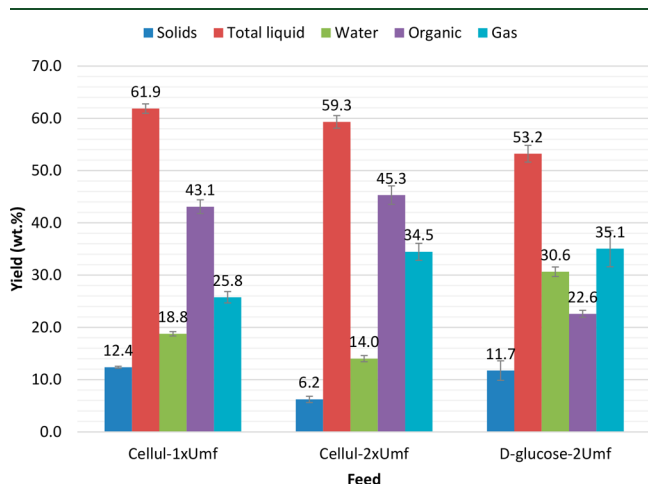


Figure 7. Pyrolysis product distribution from the pyrolysis of CE at 1xU_{mf}, 2xU_{mf}, and DG at 2xU_{mf} at 400 °C.

water than the CE and roughly the same amount of PNCGs compared to the CE. On the other hand, the DG produced half of the organic liquid yield compared to CE. These results suggested that the FRM promoted the dehydration of the DG and the polymerization of its primary pyrolysis vapors, which led to the production of char. Compared to the CE, the formation of LGA from DG was easier because it required fewer steps. However, the residence time of the LGA in the reactor was longer during the pyrolysis of DG compared to CE. Consequently, the LGA produced during DG pyrolysis was further dehydrated and polymerized into char. In fact, the longer residence time of the pyrolysis vapors in the reactor is known to promote their polymerization and charring.⁵⁵ On the other hand, the char in the reactor can catalyze the dehydration of LGA, leading to its subsequent polymerization to form secondary char.⁸⁸ In the case of CE pyrolysis, it was believed that the residence time of the LGA in the reactor was not long enough for it to undergo further dehydration and polymerization into char. Hence, LGA was collected in the biooil.

In order to prove this hypothesis, the CE was pyrolyzed at a lower fluidizing gas flow rate (1xU_{mf}) to increase the residence time of the pyrolysis vapors in the reactor from 4.7 to 5.7 s. The pyrolysis products distribution is shown in Figure 7. As the fluidizing gas flow rate decreased, the char/coke and pyrolytic water yields increased by 100.0% and 34.3%, respectively. Furthermore, the PNCG yield decreased by 25.2%. These results indicated that the longer residence time of the CE vapors in the reactor resulted in further dehydration and polymerization of the primary pyrolysis vapors into char without promoting the production of the PNCGs.

The PNCGs composition is shown in Table S4. The DG–FRM produced 62.3% more CO₂ and 17.6% more CO than the CE–FRM. This was probably due to the decomposition of LGA produced from the FRM-catalyzed pyrolysis DG. The DG–FRM produced 13.9% less hydrogen than the CE–FRM and more than 9-fold less C₂–C₆ gaseous hydrocarbons. The higher hydrogen production during the CE–FRM pyrolysis could result in higher aromatics production. This hypothesis

will be verified by biooil chemical analysis. It has been reported that during the pyrolysis of CE, the amorphous part degrades before the crystalline one. While the former produced char and PNCGs, the latter formed LGA.¹⁰⁰

3.7. Chemical Composition of the Biooils. **3.7.1. Effect of the Pyrolysis Media.** The chemical composition of the biooils from CE–sand, CE–FRM, and CE–HZSM-5 at 400 °C is shown in Table 1. In order to estimate the volatile

Table 1. Major Chemical Groups in the CE–Sand, CE–FRM, and CE–HZSM-5 Biooils

	sand	peak area (%)	
		FRM	HZSM-5
NCOs	0.07	0.44	0.00
furans	0.51	0.37	0.00
CKs	1.24	2.91	0.51
aromatics	0.13	1.38	0.89
alkene	0.00	0.04	0.00
Anh Sug	74.3	70.85	87.25
unidentified	23.74	24	11.35
total	100	100	100

fraction of the ESP biooils, the TGA of the noncatalytic and catalytic biooils mixed with methanol was conducted using a similar temperature program used for the GC/MS analysis. Briefly, the sample temperature increased from room temperature to 45 °C at 10 °C/min, and then the temperature was held at 45 °C for 10 min. Subsequently, the temperature was ramped at 3 °C/min to 250 °C and held for 5 min. The TGA curves are shown in Figure S2. At 250 °C, the solid residues of CE–sand, CE–FRM, and CE–HZSM-5 were 19.8, 16.6, and 13.6 wt %, respectively. Therefore, more than 80 wt % of the catalytic and noncatalytic ESP biooils consisted of volatile compounds that are detectable by GC/MS.

The gas chromatograms from the GC–MS analysis of the cellulose pyrolysis ESP biooil at 400 °C using sand, FRM, and HZSM-5 are shown in Figure S1. The CE–sand ESP biooil composition consisted mainly of anhydrous sugars (Anh Sug = 74.30 area %). Other groups of compounds detected were cyclic ketones (CKs = 1.24 area %), furans (0.51 area %), aromatics (0.13 area %), and noncyclic oxygenates (NCOs = 0.07 area %). The major Anh Sug was LGA (71.36 area%). The dominance of LGA in the noncatalytic biooil was in agreement with the literature.¹⁰¹

The CE–FRM sample consisted mainly of Anh Sug (70.85 area %). Other detected groups were CKs (2.91 area %), furans (0.37 area %), aromatics (1.38 area %), NCOs (0.44 area %), and alkenes (0.04 area%). The major Anh Sug was LGA (68.60 area%), and its content decreased by 3.9% compared to the CE–sand sample. The furan content decreased by 27.6% in CE–FRM catalytic biooil compared to the noncatalytic one. In contrast, the content of the aromatics, CKs, and NCO's increased by more than 10-, 2-, and 6-fold, respectively. The increase of the NCOs content in the CE–FRM sample compared to the CE–sand one was attributed to the catalytic activity of the FRM, which was reported to promote the fragmentation reactions.⁴⁴

As for the increase of the CKs content and decrease of the furan content, we speculated that the FRM promoted the conversion of some of the furanic compounds generated during the pyrolysis of CE into CKs. Omotoso et al. reported that the conversion of furfural into cyclopentanone and 2-

cyclopentenone in the vapor phase at 400 °C took place in the presence of TiO₂, and the presence of water promoted these reactions.¹⁰² The FRM catalyst contains TiO₂; furthermore, the use of FRM during the pyrolysis of CE increased the pyrolytic water content by about 2-fold compared to the noncatalytic one (Figure 3). Therefore, it was plausible to believe that TiO₂ in the FRM promoted the conversion of furans into CKs.

The aromatics detected in CE–sand and CE–FRM consisted mainly of phenols. Wodley et al. reported the presence of phenol and 2,5-dimethylphenol in the CE pyrolyzates.¹⁰³ Zhang et al. also reported the presence of phenols during the microwave-induced pyrolysis of CE at 350 °C.¹⁰⁴ The increase of the phenolic content when FRM was used could be due to the residual ash on the FRM after regeneration. Zhang et al. reported that the phosphorus-containing functional groups such as –O–P, –O–P–O–, and O=P combined with the mesopores in the biomass-derived activated carbon catalyst promoted the production of phenols during the catalytic microwave-induced pyrolysis of CE.¹⁰⁴ The FRM used during our study has been regenerated 11 times and was used previously for the pyrolysis of biomass. Hence, residual ash remained on the FRM after regeneration and sieving. It is well understood that ash from lignocellulosic biomass contains a variety of inorganic compounds such as phosphates.¹⁰⁵ Agblevor et al. reported that the phosphorus deposition on the FRM was less than 0.1 wt % and was steady after multiple regenerations of the FRM.⁴²

The CE–HZSM-5 biooil was mainly Anh Sug (87.25 area%) with LGA accounting for 82.27 area%, which was 15.3% and 19.9% higher than the CE–sand and CE–FRM samples, respectively. The aromatics content (0.89 area%) increased by more than 6-fold compared to the CE–sand sample. Phenols accounted for 32.7 area% of the aromatics generated during the CE–HZSM-5 experiment. It is worth mentioning that as the phenol content increased, furans were not detected in the CE–HZSM-5 biooil compared to the CE–sand experiment. This could be due to the fact that furans were converted into aromatics inside the HZSM-5 pores, and then some of those aromatics were further converted into phenols on the surface of the HZSM-5 as explained by Wang et al.¹⁰⁶ Mukarakate et al.¹⁰⁷ reported that phenols were produced from the catalytic pyrolysis of cellulose using HZSM-5 catalyst in the presence of steam. In our study, water was a major product during the CE–HZSM-5 pyrolysis; therefore, the phenols may have been the product of the reaction of aromatics with the pyrolytic water.

Other aromatics consisted of 1,6-dimethylnaphthalene (0.1 area%), 1H-inden-1-one, 2,3-dihydrobenzopyran-4-one (0.31 area%), and 4H-1-benzopyran-4-one (0.19 area%). The high Anh Sug content of the CE–HZSM-5 sample. The dominance of LGA in the CE–HZSM-5 sample was unexpected because most literature reported that zeolites promoted the formation of aromatics.⁶⁶ To the best of our knowledge, only one study reported that HZSM-5 promoted the formation of carbohydrates. However, they did not investigate or explain the reaction mechanism behind such a finding.¹⁰⁸

Maliekkal et al. explained the importance of the hydroxyl group of glucose in aiding LGA formation during CE pyrolysis. They explained that the breakage of the α -glycosidic bond of CE gave a glucose molecule and an epoxy end. They reported that the C4-OH group of glucose aided in the formation of LGA from the epoxide. They reported that the C4-OH group

acted as a proton shuttle between the C6-OH and the epoxide oxygen.¹⁰⁹ In a similar fashion, we speculate that the Brønsted acid sites of HZSM-5, which are known proton donors, acted as a proton shuttle and facilitated the formation of LGA during the HZSM-5-catalyzed pyrolysis of CE.

3.7.2. Effect of CE Particle Size. The chemical composition of the biooil from the FRM-catalyzed pyrolysis of CE with two different particle sizes (0.05 and 0.7–1 mm) is shown in Table 2. The major class of chemicals in the biooil from the CE700–

Table 2. Major Chemical Groups in the FRM Catalytic Biooil from CE and CE700

	peak area (%)	
	0.05 mm	0.7–1 mm
NCOs	0.44	0.56
furans	0.37	3.95
CKs	2.91	3.23
aromatics	1.38	1.24
alkene	0.04	0.00
Anh Sug	70.85	60.03
unidentified	24.00	30.98
total	100	100

FRM was Anh Sug (60.03 area%). Furans and CKs accounted for 3.95 area% and 3.23 area%, respectively. The NCOs and aromatics accounted for 0.56 area% and 0.28 area%, respectively. The LGA content decreased as the CE particle size increased from 0.05 to 0.7–1 mm. This was probably due to the fact that the LGA inside the CE700 particles spent a longer time at the reaction temperature and away from the FRM particles. Therefore, the LGA experienced further dehydration and thermal cracking to produce more stable furans, pyrolytic water, and carbon oxides. This hypothesis is corroborated by the increase of furan (Table 2), pyrolytic water (Figure 5), and carbon oxide (Table S2) content in the CE700–FRM biooil compared to the CE–FRM one.

3.7.3. Effect of the Main Oxides of the FRM on the Biooil Composition. The chemical composition of the biooils from the catalytic pyrolysis of CE using Al₂O₃ and Fe₂O₃ is shown in Table 3 along with the CE–FRM and CE–sand composition.

Table 3. Major Chemical Groups in the CE–Sand, CE–FRM, CE–Al₂O₃, and CE–Fe₂O₃

	peak area (%)			
	sand	FRM	Al ₂ O ₃	Fe ₂ O ₃
NCOs	0.07	0.44	0.26	0.00
furans	0.51	0.37	1.62	0.55
CKs	1.24	2.91	6.67	2.63
aromatics	0.13	1.38	3.72	0.5
alkene	0.00	0.04	0.00	0.00
Anh Sug	74.3	70.85	39.13	78.8
unidentified	23.74	24.00	48.6	17.51
total	100	100	100	100

Compared to the CE–sand biooil, the Al₂O₃ decreased the Anh Sug content, whereas the Fe₂O₃ increased the Anh Sug content in the biooil. The aromatic content of the biooil increased by more than 28-fold when Al₂O₃ was used and increased by more than 3-fold when Fe₂O₃ was used. Both Al₂O₃ and Fe₂O₃ had higher furan contents compared to the CE–sand sample. Compared to the CE–FRM sample, the

CE–Al₂O₃ sample had more than 4-fold higher furan content, whereas Fe₂O₃ produced only 1.5-fold more furans. The CKs content of the CE–Al₂O₃ increased by more than 2-fold, whereas the CKs content of the CE–Fe₂O₃ was slightly lower than that of the CE–FRM sample.

The effect of Al₂O₃ on the chemical composition of the CE pyrolysis biooil was in agreement with the literature. The increase of the furans content in the CE–Al₂O₃ sample was attributed to the Al₂O₃ Lewis acid sites, which are known to promote the dehydration of sugars into furans.⁹⁷ Djakaria et al. reported that during the catalytic pyrolysis of CE using Al₂O₃, the content of cyclopentanones and furans increased, whereas the content of the sugars decreased.¹¹⁰ The phenolics content of the CE–Al₂O₃ biooil was 2.33 area% compared to 0.132 area% and 1.28% in the CE–sand and CE–FRM sample, respectively, which indicated that the phenols production during the CE–FRM experiments was a combination of the ash effect and the catalytic effect of Al₂O₃.

Collard et al. investigated the pyrolysis of Fe and Ni-impregnated CE. They reported that the addition of Fe to CE promoted the formation of ketones and furans compared to the noncatalytic pyrolysis of CE. They also reported that the Fe hindered the formation of LGA, which was the opposite of what we observed in this study.⁹¹ The different effects of Fe on CE pyrolysis in our study and that of Collard et al. were attributed to their longer vapor residence time in the reactor (12 s) compared to our study (4.7 s). The longer residence time of LGA in their reactor was believed to cause its cracking into lower-molecular-weight molecules. The aromatics content of the CE–FRM sample was more than 2-fold higher than that of the CE–Fe₂O₃ sample. Additionally, the phenol content of the CE–FRM sample was 1.28 area% compared to 0.50 area% for the CE–Fe₂O₃ sample. These results supported the hypothesis of the contribution of the residual ash in the FRM in the production of phenolic compounds.

3.7.4. Comparison of the ESP Biooil from the FRM Catalyzed Pyrolysis of DG vs CE. The chemical compositions of the DG–FRM and CE–FRM are shown in Table 4. The

from DG requires a dehydration reaction, whereas depolymerization and dehydration reaction are required to produce LGA from CE. The longer residence time of the LGA in the reactor resulted in further dehydration, which produced furans. This explanation is supported by the higher pyrolytic water (Figure 7) and higher furan content (Table 4) of the DG–FRM biooil compared to the CE–FRM one. The significantly lower aromatics content in the DG–FRM biooil compared to the CE–FRM one could also be due to the longer residence time of LGA in the reactor. The longer residence time of LGA in the hot zone could have promoted its dehydration and polymerization to form char instead of liquid aromatic compounds. This explanation was corroborated by the higher char/coke content obtained from the DG–FRM (Figure 7) experiment compared to the CE–FRM one.

4. CONCLUSIONS

The catalytic pyrolysis of CE was conducted using FRM and HZSM-5 catalysts and sand as a control. Both FRM and HZSM-5 catalysts promoted the conversion of CE. HZSM-5 produced more water and PNCGs compared to the FRM and sand, whereas FRM promoted the formation of liquid organics. The FRM promoted the formation of CKs and aromatics, whereas HZSM-5 promoted the formation of Anh Sug and aromatics. The residual ash in the regenerated FRM contributed to the formation of phenols in the biooil.

The effect of CE particle size had a significant effect on the distribution of pyrolysis products distribution. The larger CE particles (CE700: 0.7–1 mm) produced higher char/coke and pyrolytic water and produced less organics and PNCGs compared to the microcrystalline CE (0.05 mm). This effect was due to the longer residence time of the primary pyrolysis vapors inside the CE700 particles, which led to further dehydration of LGA and its polymerization to form char. The CE700–FRM had higher furans and CKs and lower Anh Sug than the CE–FRM, which was due to the longer residence time of the pyrolysis vapors in the hot zone.

The major oxides of the FRM, i.e., Fe₂O₃ and Al₂O₃, produced more water and PNCGs and produced less char/coke and liquid organics compared to the noncatalytic pyrolysis of CE. Both catalysts promoted the conversion of CE into PNCGs and pyrolytic water. The Al₂O₃ promoted the formation of furans, CKs, and aromatics and hindered the Anh Sug yield. The Fe₂O₃ catalyst promoted the formation of Anh Sug and hindered the formation of aromatics.

Compared to CE, the pyrolysis of DG over FRM promoted the formation of char/coke and pyrolytic water and hindered the formation of organic liquids and PNCGs. The DG–FRM biooil had a higher furan content and lower Anh Sug content than the CE–FRM biooil. Furthermore, the aromatic content in the DG–FRM biooil was significantly lower than that in the CE–FRM biooil, which was due to the longer residence time of LGA in the reactor, leading to its further dehydration and polymerization into char.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.3c03304>.

Pyrolysis gas product composition for the noncatalytic and catalytic pyrolysis of CE in Tables S1–S4, typical chromatograms from the GC-MS analysis of the CE

Table 4. Major Chemical Compounds in the DG–FRM and CE–FRM ESP Biooils

	peak area (%)	
	DG	CE
NCOs	0.42	0.44
furans	0.78	0.37
CKs	2.40	2.91
aromatics	0.37	1.38
alkene	0.00	0.04
Anh Sug	66.95	70.85
unidentified	29.08	24.00
total	100	100

major class of compounds in the DG–FRM sample was the Anh Sug (66.95 area%). Other compounds included CKs (2.40 area%), furans (0.78 area%), aromatics (0.37 area%), and NCOs (0.42 area%). Compared to the CE–FRM, the DG–FRM had 5.5% less Anh Sug and more than 3-fold less aromatics. The major Anh Sug in the DG–FRM and CE–FRM samples was LGA, which accounted for 59.8 area% and 68.6 area%, respectively. The lower LGA content in the DG was due to its longer residence time in the reactor compared to the LGA produced from CE. In fact, the production of LGA

catalytic biooils in Figure S1, thermogravimetric analysis of GC-MS samples to estimate the nonvolatile fraction of the biooils in Figure S2, pressure drop vs superficial velocity for sand, FRM, and HZSM-5 in Figure S3, chemical composition of the ESP biooils from the CE pyrolysis experiments in Tables S5–S11 (PDF)

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Notes

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